picoRV32 Openlane2 Implementation Analysis

1. Project Objective

The primary objective of this physical design project is to achieve a target operational frequency of 200 MHz for the picoRV32 RISC-V core within a constrained die area of 350 µm × 350 µm utilizing the Openlane2 EDA flow using the Sky130 PDK.

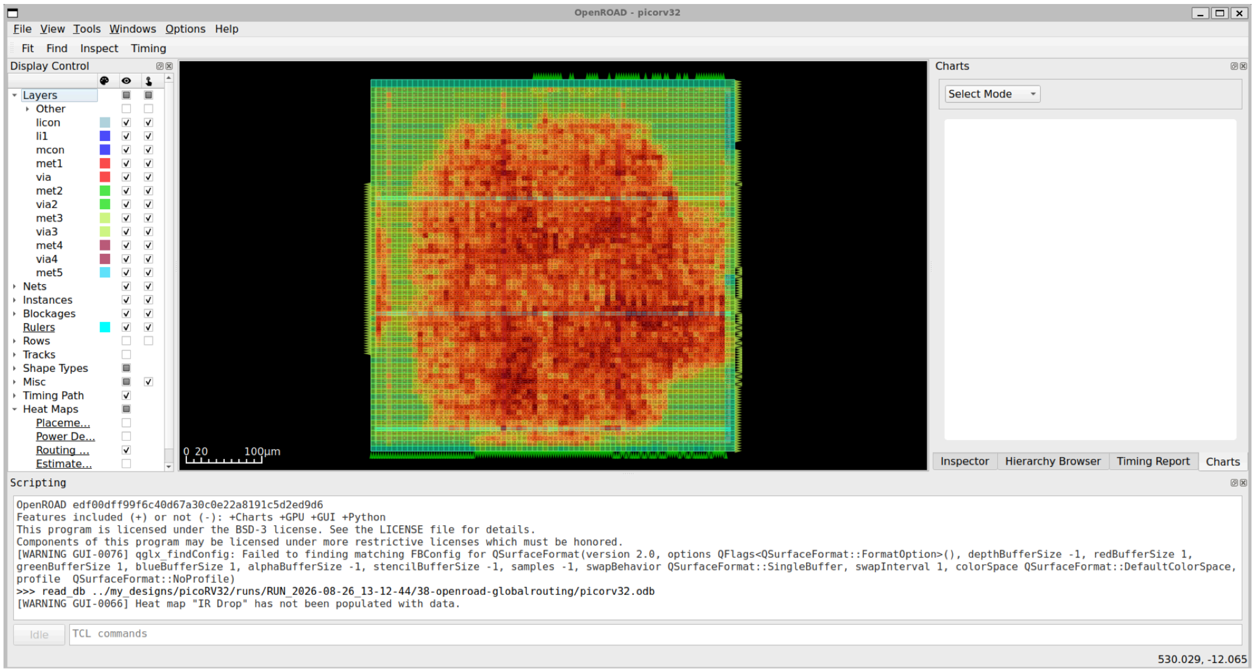
2. Development Log

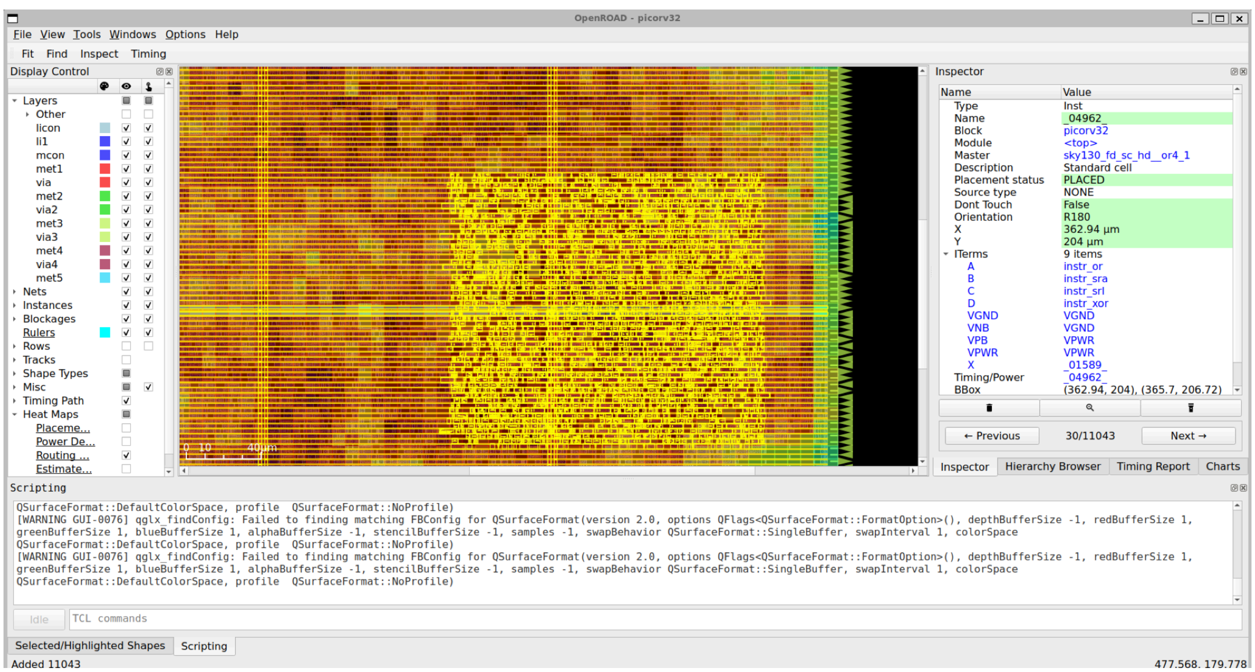
2.1 Initial Assessment

Initial physical verification revealed an XOR DRC error located on a bounding layer, which was resolved prior to detailed timing closure iterations. During place-and-route (P&R), the design failed to meet setup and hold timing across multiple Process-Voltage-Temperature (PVT) corners, preventing complete flow execution. Attempts to remediate violations via localized buffer insertion were insufficient.
To establish a stable physical implementation baseline, the target clock frequency is scaled down to 100 MHz (or 50 MHz if necessary). Subsequent optimization steps will iteratively resolve critical path delays to approach the target 200 MHz target.

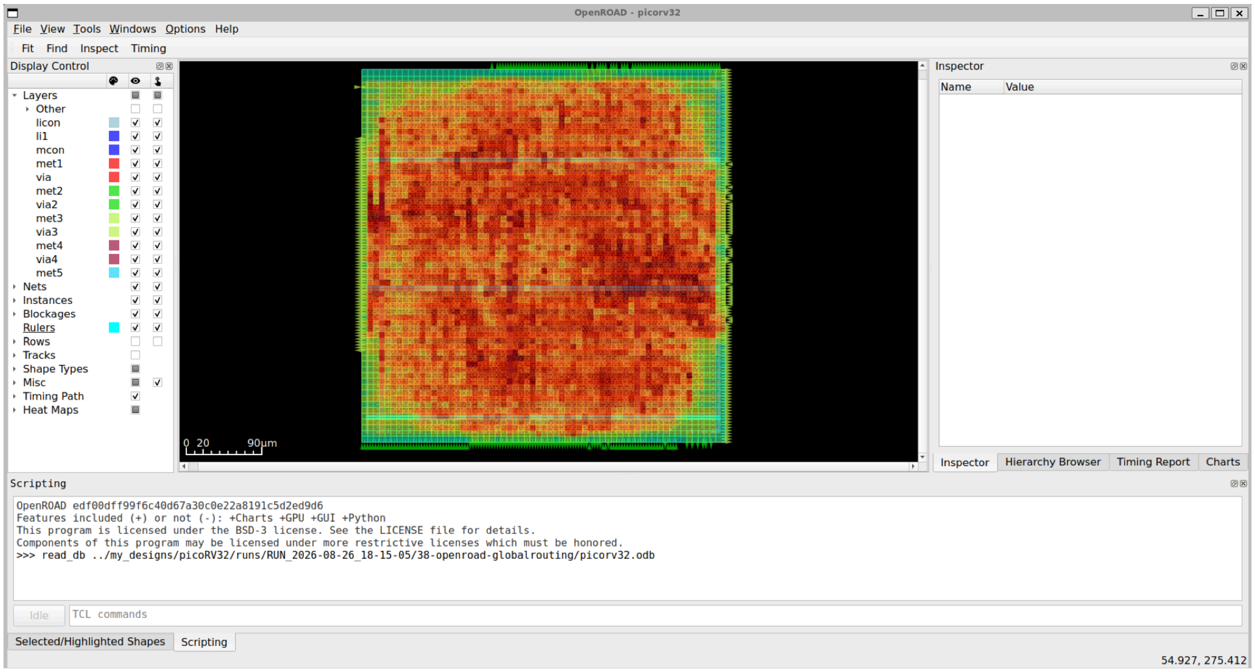
2.2 Congestion Mitigation & Fanout Optimization

Physical routing congestion was differentiated from high electrical fanout constraints to apply targeted architectural fixes. Routing congestion represents spatial metal layer saturation where routing demand exceeds available track capacity. Congested areas were identified by visualizing routing heatmaps within OpenROAD GUI.





To alleviate global routing bottlenecks across the monolithic top-level module, placement density constraints were modified to distribute standard cells more evenly across the floorplan.



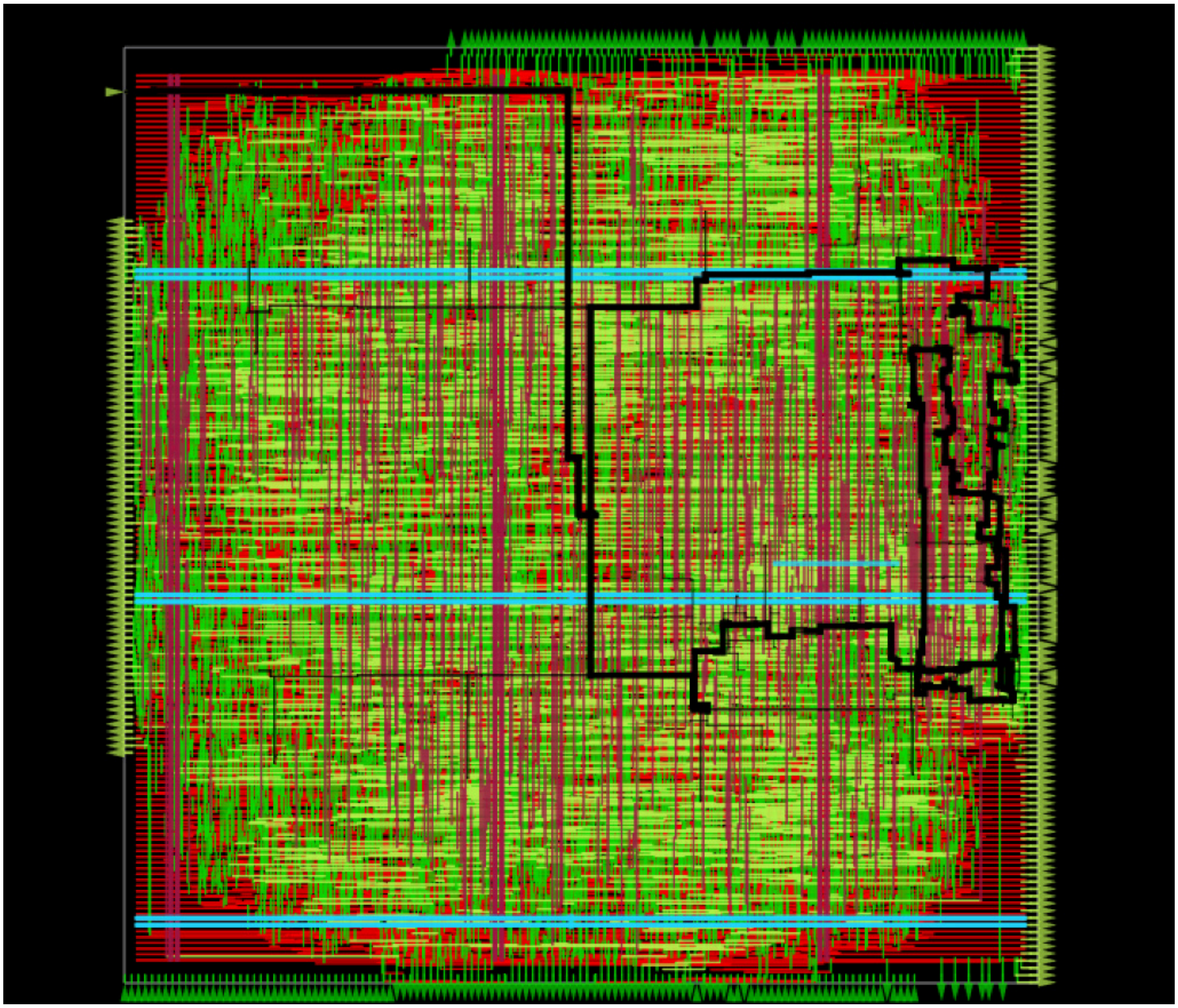
Post-adjustment analysis demonstrated a reduced congestion footprint, though localized high-density regions remained under review for timing impact.

2.3 Timing Analysis

Steps to view timing paths in OpenRoad

1. Load tech lef
2. Load lef
3. Load lib
4. Load odb
5. Load sdc
6. Load spef

Despite successful mitigation of global routing congestion, post-layout analysis indicated persistent setup timing violations. To isolate the root cause, an exhaustive evaluation of failing paths was performed. Initial analysis at the nominal corner (SS, 100°C, 1.6V) revealed a worst setup slack of -5.41 ns, confirming that logic propagation delay alone—independent of routing overhead—was insufficient to support the 200 MHz target frequency.



2.4 Synthesis Strategy Evaluation

SYNTH_STRATEGY	Gates	Area (µm²)	Worst Setup Slack (ns)	Total -ve Setup Slack (ns)
AREA 0	6706	88756.374400	-5.671488328391545	-1023.0247942784433
AREA 1	6535	87944.345600	-5.398874389252419	-2232.4491687149357
AREA 2	6780	88873.987200	-5.470457131012649	-2359.4071293417633
AREA 3	11532	109485.004800	-3.801198574606823	-239.4641068771768
DELAY 0	6918	92904.102400	-5.194763877205333	-562.1242439519766
DELAY 1	6983	93021.715200	-5.154252281980225	-381.746399962963
DELAY 2	6867	92291.014400	-4.771323027280734	-923.8622999225397
DELAY 3	6679	90728.265600	-5.488734067050145	-870.5000273669983
DELAY 4	8169	97186.960000	-5.3976957764561435	-317.8535142834082

Systematic evaluation of synthesis strategies allowed for an assessment of delay reduction potential. The data confirmed that at a 100 MHz target, logic gate propagation delays inherently result in timing failures. Consequently, achieving the 100 MHz target requires either architectural modification or frequency scaling. Subsequent analysis at a 50 MHz target frequency demonstrated positive slack across all synthesis strategies, as shown in the validation table below.

SYNTH_STRATEGY	Gates	Area (µm²)	Worst Setup Slack (ns)	Total -ve Setup Slack (ns)
AREA 0	6706	88756.374400	4.328512781831511	0.0
AREA 1	6535	87944.345600	4.601126720970637	0.0
AREA 2	6780	88873.987200	4.428065150997552	0.0
AREA 3	11532	109485.004800	6.198802535616233	0.0
DELAY 0	6918	92904.102400	4.805236344839278	0.0
DELAY 1	6983	93021.715200	4.845749716421276	0.0
DELAY 2	6867	92291.014400	3.9414161938689265	0.0
DELAY 3	6679	90728.265600	4.511266154994466	0.0
DELAY 4	8169	97186.960000	4.602304445588468	0.0

The 50 MHz target established a viable implementation baseline. This successful timing closure allows for the iterative optimization of frequency targets through subsequent timing closure and architectural refinement phases.